Max Wesley Bregler

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Education

University of California, Los Angeles (UCLA) — Los Angeles, CA

B.S. Candidate in Mathematics — *Expected Graduation: June 2029*

Relevant Coursework: Abstract Linear Algebra, Real Analysis, Numerical Analysis, Discrete Mathetics, Ordinary Differential Equations

Research

UCLA, Statistical and Relational Artificial Intelligence Lab — Los Angeles, CA

Researcher (Revisiting ImageNet) — *September 2026 – Present*

- Explored VLM (SMOL-VLM, QWEN, etc) capabilities for image recognition on MNIST and ImageNet 1k
- Compared generative and discriminative classification methods, including an attention mask for one forward pass

UCLA, BruinML Group — Los Angeles, CA

Researcher (Emergent Intelligence, 3D Reconstruction, SAR Imaging) — *January 2026 – Present*

- Discussed viability of implementing stigmergy to bandit problems to decrease environmental effects of ML
- Applied curvature consistency to loss function of 3D Reconstruction from pairs of images (under review)
- Used PDEs to extend SAR Imaging methods to nonflat surfaces by removing ground bounce

Work Experience

Golden Keys Realty Group, LLC – goldenkeyscapital.app — Elkton, MD

Software Engineering Intern — *July 2026 – September 2026*

- Developed end-to-end tenant-landlord management platform for 40+ clients over 3 properties
- Built secure authentication system with custom landlord codes for communities (Google OAuth, MongoDB)
- Engineered ticket system, tenant profile, and lease PDF uploader (FastAPI + Python)
- Connected user data to custom chatbot for tenants to gain clarity on housing issues (Gemini API)

Projects

VisitNU — San Francisco, CA

Project Lead — *January 2025 – May 2025*

- Revamped high school's campus security system to allow RFID tags to check in and out of school
- Integrated RFID tags (C & Python for backend) into VisitU (Bay Area campus security company)
- Acquired 150 student signups in 2 days, garnering over 1000 uses per week of school with positive feedback

TreeReq – treereq.com — Los Angeles, CA

Full-Stack Developer — *April 2026 – June 2026*

- Developed visual course planner for UCLA students alongside several other developers
- Implemented authentication and data management systems (Python, MongoDB)
- Integrated systems with frontend (FastAPI) and improved onboarding UI (ReactJS)

Spritide — Los Angeles, CA

Student Run Studios Developer — *January 2026 – May 2026*

- Developed topdown roguelike action RPG (Unity) with complex combat system with several other developers
- Implemented performance increases for game like object pooling hitboxes and hitscans
- Implemented core features in combat loop like critical hits and defense calculations

Skills

Languages & Tools: C, C++, C#, Python, JavaScript, PHP, TensorFlow, ReactJS, FastAPI, Remotion, MongoDB, Unity, FFMpeg, Git